

M840 Dissertation in mathematics

RG Functional Equations
&
Critical Phenomena in
Thermodynamics and Financial
Markets

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Final Dissertation for M840 TMA-04

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31 March 2026

Abstract

The paper considers the subject of critical phenomena in the physical world of thermodynamics and its analog in the economic world and the formation of bubbles in financial markets. Providing a review and comparison of a “bottom-up” approach based on statistical mechanics with a more “top-down” construction using iterative solutions to functional equations both of which exhibit the required scale invariance (power laws) and fractal qualities seen near the critical points and where the renormalization group theory provides a “bridge” between these two approaches both in a physical and mathematical sense.

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CHAPTER 1

INTRODUCTION

This paper considers two approaches to modelling different types of critical phenomena. Critical phenomena are characterised by sudden changes that span a very wide range of length or time scales of the observed feature as the critical point is approached. Well known examples in the physical world are changes in the measured specific heat of a material during phase transition [17]. Magnetic susceptibility of a ferromagnet, turbulent flow, seismic activity [40]. Such critical phenomena are also observed in the field of economics such as financial bubbles and the sudden rapid price correction at the critical point when everyone sells and brings the stock value or market index crashing down [29][33].

When these different *observables* are plotted against the relevant energy input, such as temperature, magnetic field strength or time, their graphs share common features and attributes as the critical point is approached. Depending on the values being plotted, this is typically a singularity or bifurcation. Another feature of the variables under observation, when close to the critical point are fractal like self-similarity and scale invariance, the latter being the hallmark of some underlying power law [23]. This power law is taken relative to the critical point $(X_c - X)^\beta$, where X_c represents the critical point and β represents some critical exponent. Over the top of this trend, there are higher frequency changes that might look like noise but exhibit a scale invariance. In other words, if we zoom in or out of the data, the overall features look very similar. Indeed there is a universality to these different phenomena and the exact measured values for the critical exponent β reoccur in different physical systems. Such systems are said to belong to the same universality class [28].

This paper provides a comparison of a traditional *reductionist*, “bottom-up” approach, which is physics based and a “top-down” mathematical approach, constructing solutions from recursive functional equations which provide the required self-similarity and scale invariance almost as an emergent feature. The conceptual “bridge” between these two approaches is *renormalization*, which provides descriptions of the scale-invariance and phase flows from the mathematics of dynamical systems.

Chapter 2 provides the background material for these two approaches with bottom-up approach based on a simple 2-D lattice representing the spin (magnetic dipole) of each atom, for which the Hamiltonian is calculated. The sum of the different spin interactions is then used to calculate the Gibbs free energy for the 2-D Ising Model. The subsequent solution exhibits symmetry breaking (change in entropy) and has a singularity at the critical point [1].

This model is then further refined using the Renormalisation Group (RG) in order to create the necessary bridge between the microscopic and macroscopic worlds through a process of ‘coarse-graining’ and decimation. Where the fixed points of the associated RG flow equation correspond to the critical points of the system. Inside the original RG flow construction is Schröder’s functional equation and provides the link to dynamical systems via iterated maps.

We then provide background to functional equations and their relationship with dynamical systems. Where the critical exponents are found to be the fixed points of the iterated maps and the log-periodic power-law (LPPL) is a general solution to the Schröder functional equation.

In Chapter 3 we then look at two LPPL models that have been developed to detect and model bubbles in financial market indices. The first of these is the original JLS Model developed by Sornette et al and influenced by the “bottom-up” world of statistical mechanics and thermodynamics. The second from Lynch and Mestel have taken a “top-down” approach, from consideration of Schröder’s functional equation together with the logistic equation and renormalization group.

In Chapter 4 we then provide a summary of these two approaches and conclusion in Chapter 5.

CHAPTER 2

BACKGROUND MATERIAL

This chapter provides an overview of statistical mechanics and the development of the renormalization method that has been so successful in modelling critical phenomena such as phase transition and the change in magnetic susceptibility of ferromagnet materials. Sornette [40][48][21][49] and others have used these ideas from statistical mechanics and renormalization group (RG) to model financial bubbles as critical phenomena, with the bursting of the bubble occurring at the critical point. Both these physical and financial market phenomena have features that exhibit scale invariance and self-similarity. In mathematics self-similarity is represented by a power-law. In the case of the financial modelling this is further constrained as a log power law in order to provide a discrete scale invariance (critical exponent is a complex number). We have presented Sornette's approach [49] as an example of the "bottom-up" approach due to the influence of Kadanoff's block model and Wilson's development of RG which accounted for critical phenomena in solid state physics and thermodynamics [54]. In this paper, the renormalization group approach is seen very much as a bridge between the bottom-up and top-down approaches. In particular another landmark paper written by Murray Gell-Mann & Low called Quantum Electrodynamics at Small Distances [22] also developed a renormalization group approach to deal with divergent terms in the perturbation series of the propagation functions. In doing so they recognised that the rescaling due to the renormalizability theory satisfied certain functional equations. More recently Thomas Curtright and Cosmas Zachos have revisited the RG flow equation developed by Gell-Mann & Low with a view to making more explicit the functional conjugation methods, using Schröder's functional equation to

analyse the global structure of the various RG trajectories and the interplay between continuous and discrete rescaling [13]. We have then presented Lynch and Mestel’s paper “Logistic Model for Stock Market Bubbles and Anti-Bubbles” [33], which was influenced by Curtright and Zachos work and present this as an example of the top-down characterisation. To support this, we have also included some background material on the Schröder functional equation, functional iteration and logistic equation.

2.1 Reductionist Approach (bottom-up)

Physicists have long been interested in trying to understand the properties of thermodynamics from a microscopic point of view; that is through the interaction of the atoms and molecules that make up the macroscopic material and is a subject known as Statistical Mechanics.

Statistical Mechanics proved very successful in providing models and explanations for a lot of thermodynamic properties but failed to explain physical behaviour close to the critical points that mark a change of phase.

At these critical points, the interactions between the atoms and the changes in their individual states (driven by thermal fluctuations) can extend across a range of scales from the atomic to the macroscopic.

This increase in length over which atoms start to align as the critical point is approached is known as the *correlation length*.

2.1.1 Ginzburg-Landau Theory

The Ginzburg-Landau Theory (GLT) [58] constructs a Hamiltonian for the free energy (see appendix A), based on the assumption that the divergence of the correlation length as it approaches the critical point only needs to consider the long-range fluctuations of the system and is constrained only by the rotational and

translational symmetries of the system. In GLT, the corresponding partition function becomes the functional integral

$$Z[T, \mathbf{h}] = \int D\mathbf{m}(\mathbf{x}) e^{-\beta H[\mathbf{m}, \mathbf{h}]} \quad (2.1)$$

Given the assumption that only long range fluctuations need be considered, GLT uses the following *mean-field* (saddle-point) approximation to the path integral (2.1)

$$Z[T, \mathbf{h}] \equiv e^{-\beta F[T, \mathbf{h}]} \quad (2.2)$$

where $\beta F[T, \mathbf{h}] \approx \min_{\mathbf{m}} [\beta H[\mathbf{m}, \mathbf{h}]]$

The minimum free energy occurs for the uniform vector field $\mathbf{m}(\mathbf{x}) \equiv \bar{m} \hat{\mathbf{e}}_h$ where the unit vector is aligned with magnetic field and $\bar{m}$ is the minima of the Landau free energy density.

$$f(m, h) \equiv \frac{\beta F}{V} = \frac{t}{2} m^2 + u m^4 - h m \quad (2.3)$$

2.1.2 Correlation Functions and Self-Similarity

A critical system has an additional symmetry called *dilation symmetry*. Under a change of scale, the critical correlation functions have the following functional form (see sec. 2.3.2)

$$\boxed{G_{crit}(\lambda \mathbf{x}) = \lambda^p G_{crit}(\mathbf{x})} \quad (2.4)$$

which means scale invariance a.k.a. self-similarity and therefore fractal in nature. The GLT functional integral only addresses the local symmetries of rotation and translation invariance. What it misses is the global dilation symmetry and this is what the *renormalization group* provides.

2.1.3 Ginzburg-Landau Equation

As noted in the earlier section (sec. 2.1.1), Landau mean-field approximation fails below four spatial dimensions d but is rescued by the renormalization group (RG)

theory which also goes on to account for universality. In their paper “An Introduction to the Ginzburg-Landau Theory of Phase Transitions and Non-Equilibrium Patterns” [27], Hohenberg and Krekhov propose the use of the Ginzburg-Landau theory as a unifying language to join, the otherwise disparate worlds of RG and phase transitions with patterns that are formed near linear instabilities of dynamical systems. Where a similar GLT can be derived for a non-equilibrium macroscopic phenomena. The associated real and complex Ginzburg-Landau equations provide solutions for the dynamical systems. Sornette and Johansen make use of the complex Ginzburg-Landau equation to extend their JLS model to account for the frequency shift in ω [49].

The *Ginzburg-Landau Equation* [25] is given by

$$\partial_t A = 4\partial_{xx}A + A - 3A|A|^2 \quad (2.5)$$

For Hopf bifurcations, with $\omega_* \neq 0, k_* = 0$, one substitutes $u(x, t) = \mu^{1/2}A(\mu t, \mu^{1/2}x)e^{i\omega_* t} + \text{c.c.}$ to get the complex *Ginzburg-Landau* equation

$$\partial_t A = (1 + i\alpha)\Delta_X + A - (1 + i\beta)A|A|^2 \quad (2.6)$$

These equation are best known for pattern formation (biological and chemical), with early impetus from the study of vortices in fluid dynamics and more latterly chaotic systems and the synchronisation of identical chaotic systems. Such a synchronisation corresponds to motion in an invariant subspace which is stable with respect to perturbations normal to the subspace. Any change of sign of the largest associated Lyapunov exponent (to the perturbations), with respect to the control parameter, will mark a super or sub-critical bifurcation [5].

2.1.4 Order Parameters and Response Functions

The *order parameter* is a macroscopic (thermodynamic) parameter that changes between phases and as such can be used to track phases of the material. At a phase transition, the corresponding increase in orderliness at the microscopic level is associated with *symmetry breaking*.

If we plot the spontaneous magnetization vs temperature for a ferromagnet, para-magnet or Ising model, you see a curve similar to below. There is a critical temperature T_c at which spontaneous magnetization occurs (see Figure 2.2)

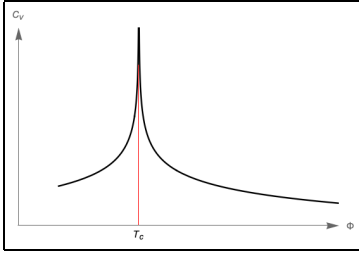


Figure 2.1: Specific Heat

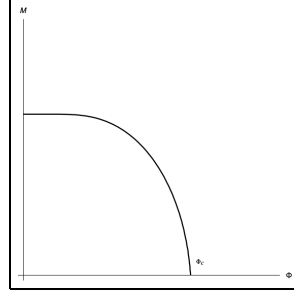


Figure 2.2: Magnetization

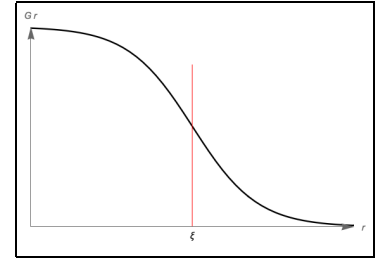


Figure 2.3: Correlation

Near T_c the correlation length as a function of T has the form of a power law

$$\xi(T) \sim (T - T_c)^{-\nu} \quad (2.7)$$

There is a similar curve for magnetic susceptibility

$$\chi \equiv \left. \frac{\partial M}{\partial H} \right|_{H=0} \quad (2.8)$$

which behaves as

$$\chi \sim (T - T_c)^{-\gamma} \quad (2.9)$$

Similarly the specific heat C_v near T_c has the same power law form

$$C_v \sim (T - T_c)^{-\alpha} \quad (2.10)$$

Indeed there a collection of thermodynamic variables (order parameters) that have response functions that follow power laws near the critical temperature T_c and associated critical components.

Critical Exponent	Quantity (Order Parameter)	Experimental Results	2D Ising Model	3D Ising Model
β	Spontaneous Magnetization (M)	0.3 - 0.38	0.125	0.31
η	Pair correlation function (G)	0 - 0.1	0.25	0.056
ν	Coherence Length	0.6 - 0.7	1.0	0.64
γ	Initial Susceptibility (χ)	1.2 - 1.4	1.75	1.25
α	Specific Heat (C_v)	0 - 0.1	0	0.12

Table 2.1: Critical Exponents and their measured values vs values from Ising Models [54]

The response functions are related to equilibrium fluctuations and their divergence imply the fluctuations are correlated over much longer distances. The experimental results in the table above show that a wide range of thermodynamic properties such as spontaneous magnetization, phase transition from liquid to gas have corresponding changes in magnetic susceptibility and specific heat capacity, respectively, that exhibit a rapid change at the critical temperature T_c , that is described by a power law. The values of these thermodynamic properties tend to either zero or infinity as the temperature $T \rightarrow T_c$.

$$\Theta_c = \frac{T - T_c}{T_c} \quad (2.11)$$

where Θ_c is referred to as the *reduced temperature*, then as $T \rightarrow T_c \Rightarrow \Theta_c \rightarrow 0$, then $F(\Theta_c)$ follows a power law (scaling behaviour).

$$F(\Theta_c) = A\Theta_c^{-n} \quad (2.12)$$

where A is a constant and n represents the critical exponent in the table above (see Table 2.1). Noting a negative $n \Rightarrow F(0) = 0$ and a positive $n \Rightarrow F(0) =$

∞ . The fascinating thing is that whilst T_c and A depend on the specific material under test, different systems (materials) can have the same values for the critical exponent, which turns out to depend only on the symmetry properties of their Hamiltonian. Systems that share a common value for the critical exponent are said to be in the same *universality class*.

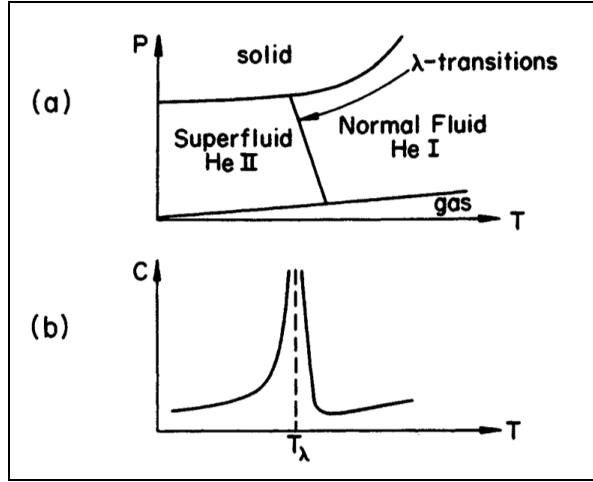


Figure 2.4: (a) Phase Diagram for He; (b) Heat capacity as a function of temperature at the λ -transition [24]

2.1.5 Kadanoff and Block Spin

The solution to the Hamiltonian for the Ising Model (see appendix B) requires consideration of a volume commensurate with the correlation length. The challenge near the critical point is of course that we have $\xi \rightarrow \infty$ making the number of calculations impossible. Kadanoff [54] described in a paper an approach to reduce the number of degrees of freedom at large correlation lengths by partitioning the spin sites into blocks and then calculating the average value for the block, thereby reducing the degrees of freedom (d.o.f.). This step can then be repeated until the number of blocks is at a manageable level with the assumption made by Kadanoff that this would reflect the scaling laws of the critical exponents. In other words when the correlation length is much larger than the spin site dis-

tances it is reasonable to assume that the four spins in each block are highly correlated and that the aggregated value of the four spin sites will also only have the two values $\{-1, 1\}$, thereby making the argument that the block of four spins can be treated the same as the original spin sites with appropriate scaling and noting that we have also effectively halved the correlation length ξ between these spin blocks.

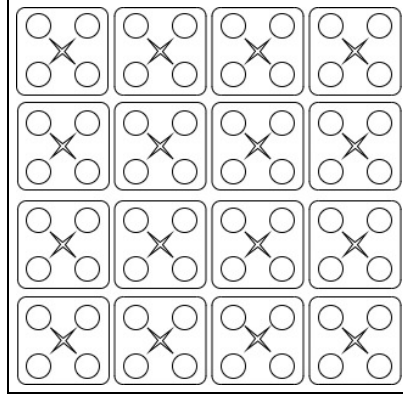


Figure 2.5: Ising spin lattice showing block spins size 4 [24]

Diagram above (Figure 2.5) provides a graphical representation of constructing the block spins. Kadanoff proposed repeating this process until correlation length for the scaled box spins is of the order $\xi \sim 1$.

The energy E/kT of the spin blocks is given by [54]

$$\sum_{n,i} K_1 s_n^{(1)} s_{n+i}^{(1)} \quad (2.13)$$

where K_1 is a constant and $s_n^{(1)}$ is the block spin variable and n enumerates the block spin site.

The original spin lattice assumed spin interaction are restricted to nearest neighbours. If these block spins are parallel (ferromagnetic) then the coupling energy is $2K$ and if anti-parallel then $-2K$.

$$\therefore K_1 = 2K \quad (2.14)$$

where $K = J/kT$. Equally the new correlation length must be half the original one

$$\xi(K_1) = \frac{1}{2}\xi(K) \quad (2.15)$$

At T_c , $K_c = J/kT_c$ and therefore $\xi(K_c) = \infty$, which is only possible if $K_1 = K_c$, leading to the nonsensical values

$$K_c = K_1 = 2K_c \Rightarrow K_c = 0 \text{ or } \infty \quad (2.16)$$

This problem arises from the assumption that the spins in the blocks are perfectly correlated. So in addition, Kadanoff also proposed that there would be an Ising interaction between the block spin, their strength given by some analytic function $f(K)$ rather than the simple doubling of $K_1 = 2K$ as shown above. This idea can be represented as the functional equation (Schröder's functional equation).

$$\boxed{\xi[f(K)] = \frac{1}{2}\xi(K)} \quad (2.17)$$

Therefore at the critical point, where we require $\xi[f(K_c)] = \infty$ we must therefore have

$$K_c = f(K_c) \quad (2.18)$$

Suppose K is near K_c , then we can write this linear approximation

$$f(K) = f(K_c) + \lambda(K - K_c), \text{ where } \lambda = \left. \frac{df}{dk} \right|_{K_c} \quad (2.19)$$

$$\therefore f(K) - K_c = \lambda(K - K_c) \quad (2.20)$$

And then using the ansatz that $\xi(K)$ obeys a power law $(K - K_c)^{-\nu}$, then we must have

$$\frac{\xi[f(K)]}{\xi(K)} = \left\{ \frac{f(K) - K_c}{K - K_c} \right\}^{-\nu} \quad (2.21)$$

but noting also that we have

$$\xi[f(K)] = \frac{1}{2}\xi(K) \quad (2.22)$$

leading to

$$\begin{aligned} \frac{1}{2} &= [\lambda]^{-\nu} \\ \therefore \nu &= \frac{\ln 2}{\ln \lambda} \end{aligned} \tag{2.23}$$

The most general form of solution for $\xi(K)$ is the log-periodic power-law

$$\boxed{\xi(K) = (K - K_c)^{-\nu} G[\ln(K - K_c)]} \tag{2.24}$$

where $G(x)$ is a periodic function of x with period λ .

2.2 Wilson's Renormalisation Group (the 'bridge')

The term “renormalization” was first introduced as a name for methods to deal with divergent series and singularities in quantum field theory [22]. Later in the 1970s, in a seminal paper by Wilson [55], the name “renormalization” was given to describe techniques for “course-graining” the microscopic statistical description of systems with many length and time scales and successfully providing a mathematical account of critical phenomena and ultimately leading to a new area of physics called statistical field theory. This so-called “course-graining” transformation consists broadly of two steps (1) a simpler description of the microscopic view (block spins) by reducing the degrees of freedom by grouping atomic elements together and averaging (see section 2.1.5) and (2) a rescaling of lengths and coupling strength K in order to stay consistent following the changes arising from step (1) [34]

Renormalization represents the macroscopic system as an aggregation of sub-systems that have been repeatedly sub-divided along some chosen geometry and iteration regime which results in a mapping from the observed macroscopic scale to a distance $|T - T_c|$ from the critical point. In the Kadanoff approach, the chosen scale is expressed by the size of the spin blocks. The major features of the RG approach is *scale invariance* and *self-similarity* of the observed elements near or

at the critical point. Following the approach from Sornette [47], giving consideration to the free energy F of the spin system. Then the renormalization approach assumes that $F(T)$ is related to another temperature T' by the following transforms

$$\boxed{\begin{aligned} x' &= \phi(x) \\ F(x) &= g(x) + \frac{1}{\mu} F[\phi(x)] \end{aligned}} \quad (2.25)$$

where $x = |T_c - T|$ and $F(x) = F(T_c) - F(T)$, such that $F = 0$ at the critical point and μ is a constant describing the rescaling of the free energy in relation to x . The function $g(x)$ represents the non-singular part of the function $F(x)$ and further assumes that $F(x)$ is continuous and $\phi(x)$ is differentiable. Then the critical points are represented by the points of singularity for the function $F(x)$, which corresponds to a finite k^{th} $d^k F(x)/dx^k$ derivative becoming infinite.

The iterative and recursive equations are given by the initial value

$$f_0(x) \equiv g(x)$$

and then

$$f_{n+1}(x) = g(x) + \frac{1}{\mu} f_n[\phi(x)], n = 0, 1, 2, \dots \quad (2.26)$$

which is the same format as the functional equations in the DFEA topics guide [35] and Sornette [46] .

$$\boxed{f_n(x) = \sum_{i=0}^n \frac{1}{\mu^{(i)}} g[\phi^{(i)}(x)], n > 0} \quad (2.27)$$

where the superscripts $^{(n)}$ represent the number of iterations i.e.

$$\begin{aligned} \phi^{(2)} &= \phi[\phi^{(1)}(x)] \\ \phi^{(n)} &= \phi[\phi^{(n-1)}(x)] \end{aligned}$$

and assuming a limit exists (see 2.3.4), we get

$$\lim_{n \rightarrow \infty} f_n(x) = F(x)$$

The equations above describe a dynamical system, where the RG flow map $\phi(x)$ captures the change in scale and the critical points correspond to the unstable fixed points of the RG flow map $\phi(x)$.

One of the challenges (before Kadanoff, Wilson and Widom), with the more traditional approach used by Landau (see Sec. 2.1.1), using function expansion and perturbation, is that whilst successful in capturing thermodynamics for the system in equilibrium and far from T_c , were nonetheless unable to account for the points of singularity at the critical points nor the scale invariance and the correlation length $\xi \rightarrow \infty$.

But the RG flow $\phi(x)$ recursive map described above produces singularities if we have the condition $|\lambda| > 1$, where λ is an eigenvalue of the RG Flow equation given by

$$\lambda = \left. \frac{d\phi}{dx} \right|_{x=\phi(x)} \quad (2.28)$$

at the unstable fixed point $\phi(x) = x$. The fixed point ensures that the same value is passed to the non-singular function $g(x)$ in the series for $f_n(x)$. In which case

$$i^{\text{th}} \text{ term of } \frac{d^k F}{dx^k} \text{ series} \propto \left(\frac{\lambda^k}{\mu} \right)^i > 1 \quad (2.29)$$

leading to a divergent series and infinite value and hence a singularity for $F(x)$.

Therefore the behaviour of the particular thermodynamic property at the critical point and its corresponding critical exponent for the power law describing the scale invariance near T_c can be deduced from the fixed points of the RG flow $\phi(x)$.

If the linearized RG map for the fixed point $\phi(0) = 0$ is taken as

$$\phi(x) = \lambda x + O(x^2) \quad (2.30)$$

then a solution for

$$F(x) = g(x) + \frac{1}{\mu} F[\phi(x)] \quad (2.31)$$

close to $x = 0$ will be described by the power law

$$\lim_{n \rightarrow \infty} f_n(x) \rightarrow F(x) \sim x^m \quad (2.32)$$

where the value of the critical exponent f_n is controlled by the two scaling factors μ and λ

$$m = \frac{\ln \mu}{\ln \lambda} \quad (2.33)$$

Re-visiting the Ising Model and expressing as a Potts Model [56], that is built up iteratively in a similar way to a Koch curve or Sierpinski triangle. In this case the construction rule is that on each iteration, we replace a single bond between spin sites with four bonds in the shape of a diamond (see Figure 2.6 below).

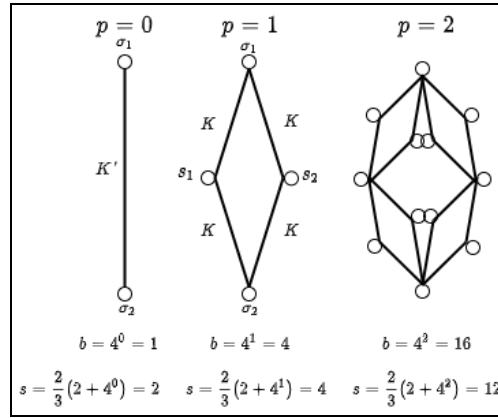


Figure 2.6: Potts Model with $Q=2$ and diamond lattice hierarchy

With each iteration we have a different scale that has a different number of bonds and spin sites

$$\begin{aligned} \#bonds &= 4^p \\ \#sites &= \frac{2}{3}(2 + 4^p) \end{aligned} \quad (2.34)$$

For a given magnification p , the energy from the interacting spin sites is given by

$$E = -J \sum_{\langle i,j \rangle} \delta(\sigma_i, \sigma_j) \quad (2.35)$$

where J is the coupling term and the sum is taken over nearest neighbours. The partition function provides a full description of the system state for a particular configuration

$$Z(\beta) = \sum_E e^{-\beta E} \quad (2.36)$$

As previously noted, at the microscopic scale, the calculation for Z would be impossible. However recasting the Ising Model lattice sites to the fractal design above, we can construct an exact RG equation and solve the problem exactly. This is equivalent to the 'decimation' of the number of spin sites by Kadanoff's spin blocks. For $p = 1$ we have

$$K^{\delta(\sigma_1 s_1) + \delta(\sigma_1 s_2) + \delta(\sigma_2, s_1) + \delta(\sigma_2 s_2)} \quad (2.37)$$

where $K = e^{-\beta J}$, we can then calculate the partition summation using s_1 and s_2 as indices

$$\sum_{s_1, s_2} K^{\delta(\sigma_1 s_1) + \delta(\sigma_1 s_2) + \delta(\sigma_2, s_1) + \delta(\sigma_2 s_2)} \quad (2.38)$$

With the final result at this level depending on if we have $\sigma_1 = \sigma_2$ or $\sigma_1 \neq \sigma_2$, we have the two different sums [14].

For $\sigma_1 \neq \sigma_2$, we have

$$(2K + Q - 2)^2 \quad (2.39)$$

and for $\sigma_1 = \sigma_2$

$$(K2 + Q - 1)^2 \quad (2.40)$$

where $Q = 0, 1, 2, \dots$ is the Potts number [56] and for the 2-D Ising Model $Q = 2$ and these two sums can be combined to give

$$(2K + Q - 2)^2 \left[1 + \left(\frac{(K^2 + Q - 1)^2}{(2K + Q - 2)^2} - 1 \right) \delta(\sigma_1, \sigma_2) \right] \quad (2.41)$$

and restating as

$$(2K + Q - 2)^2 K'^{\delta(\sigma_1, \sigma_2)} \quad (2.42)$$

and using the identity

$$K'^{\delta(\sigma_1, \sigma_2)} = 1 + (K' - 1)\delta(\sigma_1, \sigma_2) \quad (2.43)$$

where

$$K' \equiv \left(\frac{K^2 + Q - 1}{2K + Q - 2} \right)^2 \quad (2.44)$$

For a regime where only the interaction coupling J is considered for nearest neighbours, this recursion relates spin state energy at magnification scale p with the previous scale $p - 1$, with a scale dependent coupling K_p via the RG map [15]

$$K_{p-1} = \phi(K_p) \equiv \left(\frac{K_p^2 + Q - 1}{2K_p + Q - 2} \right)^2 \quad (2.45)$$

where the coupling parameter $|K - K_c|$ serves as the control parameter x .

$$Z_p(K) = ((2K + Q - 2)^2)^{4p} Z_{p-1}[\phi(K)] \quad (2.46)$$

which allows us to restate the free energy at magnification p in terms of the coupling K

$$\begin{aligned} f_p(K) &= \frac{1}{4^{p+1}} \ln Z_p(K) \\ &= \frac{1}{4^{p+1}} \left(\ln \left[((2K + Q - 2)^2)^{4p} Z_{p-1}[\phi(K)] \right] \right) \\ &= \frac{1}{4^{p+1}} (2(4^p) \ln(2K + Q - 2) + \ln(Z_{p-1}[\phi(K)])) \\ &= \frac{2(4^p)}{4^{p+1}} \ln(2K + Q - 2) + \frac{1}{4^{p+1}} \ln(Z_{p-1}[\phi(K)]) \\ f_p(K) &= g(K) + \frac{1}{4} f_{p-1}(K') \end{aligned} \quad (2.47)$$

which has the original form, with $\mu = 4$

$$\boxed{f(K) = g(K) + \frac{1}{\mu} f(K')} \quad (2.48)$$

and

$$g(K) = \frac{1}{2} \ln(2K + Q - 2) \quad (2.49)$$

as required.

2.3 Holistic Approach (top-down)

The following sections provide an overview of some aspects of functional equations relevant to their description of dynamical systems.

2.3.1 Background on Functional Equations, Iteration and Fixed Points

A fixed point x^* of a function $g(x)$ is defined by the property $g(x^*) = x^*$. One way to try and find a fixed point (there may be more than one depending on $g(x)$) is repeatedly call the function, starting with x_0 and then subsequently passing the function's output back in to the function.

$$\begin{aligned} x_1 &= g(x_0) \\ x_{n+1} &= g(x_n), \quad n = 1, 2, 3, \dots \end{aligned}$$

Stated another way, this leads to functional composition

$$\begin{aligned} x_1 &= g(x_0) \\ x_2 &= g(g(x_0)) = g^{(1)}(x_0) \\ x_{n+1} &= g(g^{n-1}(g^{n-2}(\dots g(x_0)\dots))) = g^{(n)}(x_0) \end{aligned}$$

Clearly the question of whether or not this will converge to a fixed point (and therefore a solution) will depend on $g(\cdot)$ and the starting point x_0 . For there to be an attracting fixed point in the interval $[a, b]$ then g must be a continuous function with the the following properties [16]

$$g : [a, b] \rightarrow [a, b] \text{ and } |g'(x)| < 1 \quad (2.50)$$

An alternative approach is the Newton, Newton-Raphson method

$$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)}, \quad n = 0, 1, 2, 3, \dots \quad (2.51)$$

In the early 1870s, Ernst Schröder was trying to understand Newton's method in the complex plane [41] who discovered the equation (that now bears his name)

$$\boxed{f \circ \varphi = \lambda f} \quad (2.52)$$

where φ is a given quantity and holomorphic selfmap of the unit disc $U = \{z \in \mathbf{C} : |z| < 1\}$ with the goal of finding $\lambda \in \mathbf{C}$ and f holomorphic on U . Schröder's equation is the eigenvalue equation for the composition operator C_φ which is defined as $C_\varphi f = f \circ \varphi$. Whilst he was unsuccessful in developing this into a general theory, he was the first to use conjugation as means to understanding iteration near an attractive fixed point.

2.3.2 Scaling and Scale Invariance

A function is homogeneous of degree n if it has the functional form

$$\boxed{f(\lambda x, \lambda y, \lambda z, \dots) \equiv \lambda^n f(x, y, z, \dots)} \quad (2.53)$$

In thermodynamics, if the scale of the system is increased by a factor of λ with no change in its intensive¹ properties, then all its extensive² properties are increased by the scalar factor λ [53].

In Thermodynamics, near the critical point T_c , the magnetic field H , magnetization M and reduced temperature $t = (T - T_c)/T_c$ are related by

$$H = M |M|^{\delta-1} j\left(\frac{t}{|M|^{1/\beta}}\right) \quad (2.54)$$

where $j(x)$ is the “scaling” function and β and δ are two critical point exponents. Where by convention β is the critical exponent for magnetization M , δ is the critical exponent for the magnetic field H , γ is the critical exponent for susceptibility given by

$$\gamma = \beta(\delta - 1) \quad (2.55)$$

and α is the critical exponent for specific heat C_H given by

$$\alpha + 2\beta + \gamma = 2. \quad \text{Rushbrooke's scaling law} \quad (2.56)$$

¹intensive properties do not depend on the quantity of matter e.g. pressure, temperature

²extensive properties do depend on the quantity of the matter e.g. entropy, energy, volume

When $\alpha = 0$ giving $2\beta + \gamma = 2$ generally means a logarithmic rather than a power law divergence [53]. The critical exponents can be expressed as functions of ν , η and d

$$\begin{aligned}
 \alpha &= 2 - \nu d && \text{Josephson's scaling law} \\
 \beta &= \nu \frac{d - 2\eta}{2} \\
 \gamma &= \nu(2 - \eta) && \text{Fisher's scaling law} \\
 \delta &= \frac{d + 2 - \eta}{d - 2 + \eta}
 \end{aligned} \tag{2.57}$$

where ν is called the *correlation length exponent*, η is called the *anomalous dimension* and d is the *dimensionality*, which is crucial for explaining systems near their critical points. The presence of the anomalous dimension is central to the theory of critical phenomena, which gives the remarkable result that near the points of criticality, microscopic scale lengths are important to macroscopic behaviour (contrary to Landau's mean-field theory assumptions). What this means is, for some quantity F (e.g. free energy) that depends on the microscopic measure l (Ising model lattice spacing - see appendix B) and the correlation length ξ then it is only legitimate to replace l/ξ with 0 in the function $F(l/\xi)$, if $F(x)$ is not singular in the limit $x \rightarrow 0$. This gives three possibilities [24]

1. $F(x) \rightarrow 0$ as $x \rightarrow 0$
2. $F(x) \sim x^{-\sigma}\Phi(x)$ as $x \rightarrow 0$ with $\sigma > 0$ and $\Phi(x)$ regular in the limit $x \rightarrow 0$
3. None of the above

The answer is (2) for critical phenomena and the reason why renormalization group scaling is required to replace the more usual dimensional analysis used in physics.

2.3.3 Conjugacy Equations

Chapter 8 of Kuczma et al Iterative Functional Equations [31] introduces the conjugacy equation as

$$\boxed{\varphi(f(x)) = g(\varphi(x))} \quad (2.58)$$

The mappings $f : X \rightarrow X$ and $g : Y \rightarrow Y$ are said to be *conjugate* iff there exists a bijective solution $\varphi : X \rightarrow Y$ for equation (2.58). If g is linear, then the procedure of transforming f into g is known as *linearization*, which often plays an important role in the study of dynamical systems. For example, a dynamical system that is described by a non-linear differential equation such as

$$\dot{x} = F(x), \quad x \in \mathbb{R}^N$$

can be replaced by the linear equation

$$\dot{z} = F'(0)z$$

so long as F meets certain criteria.

2.3.4 Embeddings in Iteration Theory

The following section is taken from Marek Cezary Zdun's survey paper on selected topics of embeddability of certain mappings [57]. These embeddings are at the heart of the Renormalization Group flow used by Gunn-Murray and Low [22] to manage divergent series in quantum field theory. Wilson's treatment of Kadanoff's block Ising model [54], to model the critical points in solid state physics and Lynch and Mestel's [33] use of the logistic mapping to derive the log periodic power laws for financial bubbles via Schröder's functional equation.

Let X be a real manifold and $f : X \rightarrow X$ be a given continuous injection. A family of homeomorphisms $f^t : X \rightarrow X, t \in \mathbb{R}$, such that

$$f^t \circ f^s = f^{t+s}$$

is said to be a *flow* or an *iteration group*. If in addition, $f^1 = f$ then the flow is said to be an *embedding* of f or an *iteration group* of f [57]. In the context of dynamical systems, the set X is said to be a phase space. The embeddability of a given mapping of an iteration is the same as an extension of a discrete time dynamical system to continuous time dynamical system without changing the phase space. In 1884, G Koenigs proved a fundamental theorem [30] on the existence and uniqueness of analytic solutions for Schröder's functional equation

$$\varphi(f(x)) = s \varphi(x) \quad (2.59)$$

which proved that a formal solution of Schröder's equation is predicated on the existence of the limit

$$\varphi(x) = \lim_{n \rightarrow \infty} \varphi(x) = \lim_{n \rightarrow \infty} \frac{f^n(x)}{s^n} \quad (2.60)$$

By doing this Koenig was the first to consider the problem of the embeddability for the holomorphic functions and more recently considered by Thomas Curtright and Cosmas Zachos (see 2.3.6), where they use it to analyze the global structure of various renormalization group trajectories [13].

2.3.5 Dynamical Functional Equations and Logisitic Equation

The subject of functional equations is broadly a search for functions that are solutions that satisfy a functional template, which acts to constrain the relationship in some way [43]. There are many types of functional equations that have been studied, over hundreds of years [3][19][31][35]. Simple examples are

$$\begin{aligned} f(x+y) &= f(x)f(y) && \text{satisfied by exponential functions} \\ f(xy) &= f(x) + f(y) && \text{satisfied by logarithmic functions} \end{aligned}$$

to complex quantum field theory and statistical mechanics such as the Yang-Baxter equation [19].

$$\sum_{\alpha, \beta, \gamma=1}^N R_{jk}^{\alpha\beta}(x-y) R_{i\alpha}^{l\gamma}(x) R_{\gamma\beta}^{mn} = \sum_{\alpha, \beta, \gamma=1}^N R_{ij}^{\alpha\beta}(y) R_{\beta k}^{\gamma n}(x) R_{\alpha\gamma}^{lm}(x-y) g$$

Dynamical functional equations involve the functional composition of one or more functions within the functional equation. This composition corresponds to discrete-time dynamics via iteration [35] (see sec. 2.3.1).

2.3.5.1 Linear Functional Equations

Linear functional equations are of the type [44] [31]

$$f(g(t)) = h(t)f(t) + F(t) \quad (2.61)$$

where $g(t)$, $h(t)$ and $F(t)$ are given and the function $f(t)$ is unknown. An example of first order linear dynamical functional equation is [35].

$$\boxed{f(x) = \mu^{-1}f(\phi(x)) + g(x)} \quad (2.62)$$

2.3.5.2 Schröder's Functional Equation

This equation is referenced throughout the paper and forms a central part of providing a solution to the RG flow and its definition is [35]

$$\boxed{\psi(\phi(x)) = \lambda\psi(x) + g(x)} \quad (2.63)$$

where μ and λ are non-zero constants and $g(x)$ is a known fixed function and alternatively expressed with the linear operator T

$$T(f)(x) = \mu^{-1}f(\lambda x) \quad (2.64)$$

which can be used to construct a recurrence relation which is iterated to provide a map that scales.

Kuczma [31] refers Schröder's equation to as the fundamental equation of linearization .

$$\psi(\phi(x)) = S \Lambda(x) \quad (2.65)$$

where $\Lambda \in \mathbb{K}^{N \times N}$ where $\mathbb{K}$ stands for either the reals $\mathbb{R}$ or complex $\mathbb{C}$. If the function $\phi(x)$ has a fixed point at the origin and is differentiable there, then by the conjugacy equation (2.58), equation (2.65) can have a smooth solution only if the matrices Λ and $\phi'(0)$ are conjugate: $\phi'(0) = C\Lambda C^{-1}$

where $C = \psi'(0)$ (Ch8.2 [31]).

2.3.5.3 Logistic Equation

The logistic equation is a well known and much studied iterative equation within the field of dynamical systems and defined as

$$x_{t+1} = \mu f^{(t)}(x)(1 - x_t), \text{ for } t = 0, 1, 2, \dots, n \quad (2.66)$$

and despite its simplicity, repeated iterations produce complex output as μ is varied, revealing bifurcation and period doubling of its fixed points (Fig. 2.7). The iteration subscript t can be interpreted as a discrete time step for the evolution of x_t produced from the iterative composition of $f(x)$.

$$\begin{aligned} f^{(0)}(x_0) &= x_0 \\ f^{(1)} &= \mu f^{(0)}(x_0)(1 - x_0) \\ f^{t+1}(x) &= \mu f^{(t)}(x_t)(1 - x_t) \end{aligned} \quad (2.67)$$

Ermst Schröder found a closed form solution for all values of the iterate t [52].

$$g_t(x) = \sin^2(2^t \arcsin(\sqrt{x})) \quad (2.68)$$

which provides a continuous form of the discrete map iteration that produces the well known bifurcation diagram (Fig 2.7). This provides the same discrete to continuous mapping employed by Curtright and Zachos (Sec. 2.3.6) [13] and Lynch and Mestel [33] (Sec. 3.2).

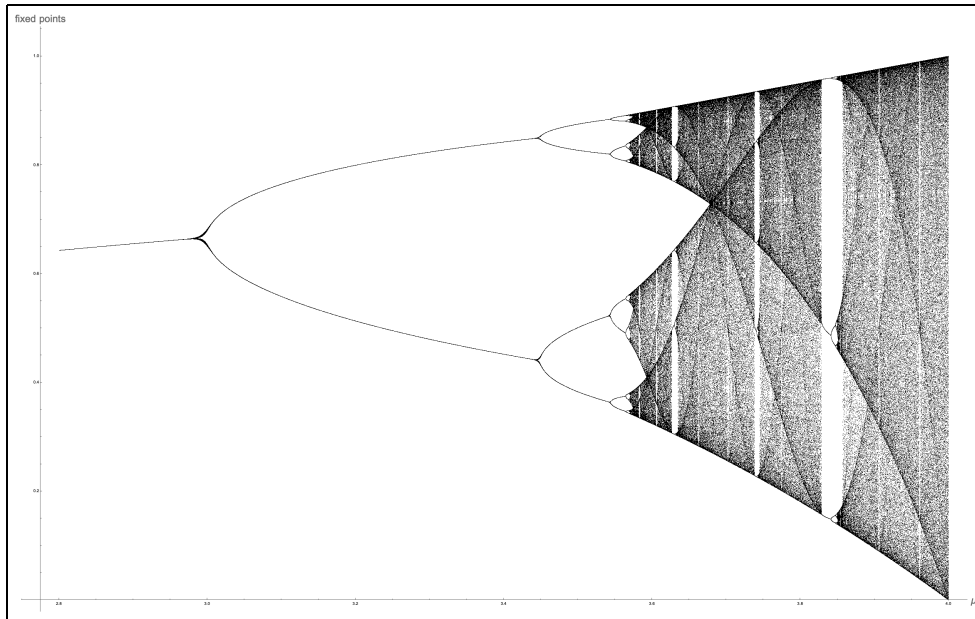


Figure 2.7: Logistic Bifurcation Diagram

2.3.6 Continuous time trajectories from discrete time iterative mapping

In their paper “Evolution Profiles and Functional Equations” [12], Curtright and Zachos illustrate the use of Schröder’s functional equation and its auxiliary equation to produce continuous trajectories from the iteration of discrete time mappings (orbits). An approach they use in another of their papers “Renormalization Group Functional Equations” [13] and in turn is used by Lynch and Mestel in their paper “Logistic Model for Stock Market Bubbles and Anti-bubbles” [33].

From [12] consider the differential equation

$$\frac{dx(t)}{dt} = v(x(t))$$

which represents a time-translation invariant law such as conservation of energy. This can then be integrated, with an initial value $x(0)$ to obtain the trajectory as

a family of functions indexed by the time parameter t .

$$x(t) = f_t(x(0))$$

Then for any given time, t may be scaled to consider the unit time increment $\Delta t = 1$ such that

$$x(1) = f_1(x(0))$$

but then for time-translation invariant systems, further evolutionary time steps follow as

$$x(t+1) = f_{t+1}(x(0)) = f_1(x(t)) \quad (2.69)$$

Given the above regime, values can be obtained for points on an integral lattice e.g. $t = \dots, -2, -1, 0, 1, 2, \dots$ etc by appropriate number of iterations to obtain the *splinter* of x .

$$x(n) = f_1(f_1 \dots (f_1(x) \dots))$$

But instead, identify an alternative approach that provides all iterates of $f_1(x)$, providing a continuous time evolution by using Schröder's construction of f_t . Recasting equation (2.59)

$$s\Psi(x) = \Psi(f_1(x)), \quad s \neq 1 \quad (2.70)$$

with the origin a fixed point of $f_1(0) = 0$, then $\Psi(0) = 0$ and $\Psi'(0) \neq 0$. The inverse function Ψ^{-1} is given by Poincaré's equation

$$\Psi^{-1}(sx) = f_1(\Psi^{-1}(x)) \quad (2.71)$$

Then by solving Schröder's auxiliary equation Ψ and its inverse Ψ^{-1} , the full continuous trajectory, for a suitable domain, around the fixed point, is given by the equation.

$$\boxed{x(t) \equiv f_t(x) = \Psi^{-1}(s^t \Psi(x))} \quad (2.72)$$

One other condition is that in the limit $s \rightarrow 1$ all iterates degenerate to the identity map $f_0(x) = x$, where $s = f'_1(0)$. In the case where $s = 1$, then replace $f_1(x)$ with $sf_1(x)$ in Schröder's equation and take the limit $s \rightarrow 1$ only at the end of the calculations. In general a formal solution is dependent on Koenigs function (see eqn 2.60)

2.3.7 Renormalization Group and Functional Equations

As noted previously, the term “renormalization” was coined in the world of quantum field theory but earlier examples can also be found in Weiss theory of ferromagnetism and the idea of turbulence diffusivity introduced by Boussinesq [34]. All the above exhibit typical features of a renormalization process such as randomness, physical processes that operate over varying distance and timescales, scaling constants and associated renormalized quantities of the 'bare' measurements. These features are also features of chaotic or fractal systems.

In their paper [13] Curtright and Zachos take the approach of using functional equations to analyse the global structure of the RG flow. The approach they take is to recast the renormalization group (RG) in terms of the functional conjugation methods of Schröder [31] which provides a description of the physical systems under both very small and very large changes in scale.

From [13], we have a discrete change in length scale for the coupling u given by $u \mapsto \sigma(u)$ which is a “step-scaling” function often deployed in monte-carlo simulations; where $\sigma(u)$ represents a discrete Δt sampling of the renormalization trajectory. The RG trajectory has a continuous evolution under changes in the $\log$ of the length scale and the initial state is given by $u \equiv u(t)|_{t=0}$. It is assumed the trajectory flow is abelian under group composition e.g. $\sigma(u) = u(t)|_{t=1}$, $\sigma(\sigma(u))|_{t=2}$ etc. In RG theory, the equation describing the local flow is

$$\frac{du}{dt} = \beta(u) \quad \text{where } \beta \text{ is the Callan-Symanzik function} \quad (2.73)$$

which is equivalent to

$$(\ln \lambda) \frac{\Psi(u)}{\Psi'(u)} \quad \text{where } \Psi \text{ is Schröder's functional equation} \quad (2.74)$$

both of which appear in Gell-Man and Low's paper "Quantum Electrodynamics at Small Distances" [22]. It is implicit, that the above equations are time translation invariant and hold more generally for $u(t)$ giving $\beta(u) \rightarrow \beta(u(t))$ and $\Psi(u) \rightarrow \Psi(u(t))$. From (2.73) and (2.74) we then get

$$\frac{d \ln \Psi(u)}{du} = \frac{\ln \lambda}{\beta(u)} \quad (2.75)$$

and the definite integral provides the total change in Ψ in respect to a finite change in the coupling u

$$\Psi(u_2) = \lambda^{\int_{u_1}^{u_2} \frac{du}{\beta(u)}} \Psi(u_1) \quad (2.76)$$

which can be expressed in terms of the evolution of the Schröder's equation under the coupling $u \rightarrow u(t)$ as

$$\Psi(u(t)) = \lambda^t \Psi(u) \quad (2.77)$$

and then taking the inverse of both sides, provides the *global flow* in [22] by

$$u(t) = \Psi^{-1}(\lambda^t \Psi(u)) \quad \text{where } 1/\ln \lambda \text{ sets the scale of } t \quad (2.78)$$

as before (see Sec. 2.3.6).

CHAPTER 3

APPLICATION OF MODELS

3.1 Financial Bubbles (bottom-up)

Various researchers have taken the ideas of critical phenomena and statistical mechanics and looked at applying them to aspects of financial markets. Specifically, making the argument that financial bubbles and the subsequent crash (or market correction) is a critical event that may be modelled and explained using renormalization to bridge the activities of traders (microscopic view) to the creation of the bubble and crash (macroscopic view). Chief amongst the researchers in this field is Didier Sornette [29][48][45].

In the paper *Crashes as Critical Points* [29], Joansen, Ledoit & Sornette put forward the hypothesis that stock market crashes are the result of long range correlations between traders causing a bubble. At some critical point, there is a drastic change in trader sentiment that then leads to a rapid sell-off, causing the market to crash. The transition from the exponential growth in the price of the securities, during the bubble to the crash is discontinuous and again similar to the singularities seen at critical points such as magnetization or a change in state in some physical material.

In [50], Sornette et al had identified the growth in the market index, during the bubble phase followed a log-periodic power-law (LPPL). This is similar to the power laws describing behaviour of thermodynamic observables as they approach the critical point.

$$F_{lp}(\tau) = A_2 + B_2 (\tau_c - \tau)^{m_2} [1 + C \cos(\omega \log(\tau_c - \tau))] \quad (3.1)$$

Where the log periodic power function F_{lp} is a function of $\tau = t/T$, which is the time in units of T and τ_c is the time of the crash (critical event) and m_2 is the critical exponent.

What follows is a description of the so called Johansen-Ledoit-Sornette (JLS) Model [29]. In order to give an explanation to the observed log periodic power law (LPPL), the authors propose a simple equity stock that does not pay a dividend. The traders purchasing the stock are purely speculative, in other words they are looking to make a profit in the short term by buying the stock and then selling once they have locked in a sufficient profit. To further simplify, the model also ignores factors such as interest rates, risk appetite of individual traders and the market is assumed to be very liquid and no friction in selling stock. The model further assumes the efficient market hypothesis that states all information on the stock is already priced in to the stock's value and no arbitrage opportunities. Mathematically this is often expressed as the martingale hypothesis

$$\forall t' \quad E_t[p(t')] = p(t) \quad (3.2)$$

where $p(t)$ denotes the price of the stock at time t and $E_t[\cdot]$ represents the conditional expectation given all previous information up to time t . Therefore since there is no dividend payment, its fundamental value is $p(t) = 0$. Or in words there is nothing to make the investor think there is a positive return on holding the stock. Therefore, by this definition, if $p(t)$ becomes positive, then this indicates a speculative bubble (within the context of the efficient market hypothesis).

The authors then look to model the likelihood of a crash occurring at time t using a hazard rate $h(t)$ that is defined as

$$h(t) = \frac{q(t)}{1 - Q(t)} \quad (3.3)$$

where $Q(t)$ is the cumulative distribution function (cdf) of the time of the crash, $q(t) = dQ/dt$ is the probability density function (pdf). The hazard rate is a concept from reliability analysis and used in predicting the probability of failure at any instant in time. In this case it is the probability per unit time that the crash

will happen in the next instant. In addition, a jump process j captures the discontinuity in $p(t)$ at the time the crash happens and has the value zero before the crash and one afterwards. The probability that a crash will occur at $t + dt$ is given by $h(t)dt$ conditioned on the crash not having yet occurred. Therefore the expectation of dj having occurred is given by

$$E_t[dj] = 1 \times h(t)dt + 0 \times (1 - h(t)dt) = h(t)dt \quad (3.4)$$

then the asset's price dynamics before the crash is given by

$$dp = \mu(t)p(t)dt - \kappa p(t)dj + \sigma(t)p(t)dW \quad (3.5)$$

where $\mu(t)$ is the return on the asset, $\sigma(t)$ is the volatility of the asset's price, dW is the infinitesimal increment of a standard Weiner process with a mean of zero and a variance equal to dt and $\kappa \in (0, 1)$ represents the percentage price drop of asset post crash. Then the time dependent expectation of dp is given by

$$\begin{aligned} E_t[dp] &= \mu(t)p(t)dt - \kappa p(t)E_t[dj] + \sigma(t)p(t)E[dW] \\ &= \mu(t)p(t)dt - \kappa p(t)h(t)dt \end{aligned} \quad (3.6)$$

the martingale hypothesis means we must have $E_t[dp] = 0$ and $E_t[dW] = 0$

$$\therefore \mu(t)p(t)dt - \kappa p(t)h(t)dt = 0 \quad (3.7)$$

$$\therefore \mu(t) = \kappa h(t) \quad (3.8)$$

which shows that the price increase is driven by the probability of a crash and the argument goes this is the increased risk/reward for the imitation traders to continue entering the market and buying the assets.

Which gives the following equation before the crash ($\kappa p(t)dj = 0$)

$$\frac{dp}{p(t)} = \kappa h(t)dt + \sigma(t)dW \quad (3.9)$$

integrating from time t_0 gives

$$\begin{aligned} \int_{t_0}^t \frac{dp}{p(t')} &= \int_{t_0}^t \kappa h(t')dt' \\ \therefore \log \left[\frac{p(t)}{p(t_0)} \right] &= \kappa \int_{t_0}^t h(t')dt' \end{aligned} \quad (3.10)$$

which shows that as the likelihood of the crash increases, then the faster the price must increase in order to reward the buyer.

Clearly this continued price increase cannot continue indefinitely and something happens within the trading community that triggers a huge sale of stock that leads to the market index crashing. The model is based on the hypothesis that all the traders can be represented by a network (eg family, friends, colleagues etc) in which they all influence one another locally; where a trader will be directly connected to k nearest neighbours. The trader can be influenced from two different sources, (a) the opinions of his k nearest neighbours in a positive way i.e. they will look to copy their investment decisions and (b) some external signal that only the trader receives and could be a contrarian signal.

The JLS model proposed a model in which each trader can be in one of two states s_i , namely *buy* $s_i = +1$ or *sell* $s_i = -1$. The state of the trader i at a given time t is then modelled with the Markov process:

$$s_i = \text{sign} \left(K \sum_{K \in N(i)} s_j + \sigma \epsilon_i + G \right) \quad (3.11)$$

where $\text{sign}(\ast)$ function returns $+1$ for a positive sum and -1 for a negative sum and K is a positive constant that denotes the coupling strength between the traders i.e. how strongly they are influenced. $N(i)$ is the number of traders who influence trader i , s_j is the current state of the trader and σ is the tendency to act upon an external signal and ϵ_i is random variable with a standard normal distribution. The G term represents an external global influence, where $G > 0 \Rightarrow s_i = +1$, $G < 0 \Rightarrow s_i = -1$.

The relative strengths of K and σ determines the level of imitation between the traders and therefore how aligned their buy/sell positions are. At the time of the crash then K will have the critical value of K_c and all the traders will be either $+1$ or -1 .

The likelihood that a large group of traders suddenly find themselves taking the

same view of the market and therefore their stock position, is called the susceptibility of the system. Analogous to the idea of magnetic susceptibility in solid state physics (see appendix A), this refers to how susceptible the global set of traders are to external information triggering a common response.

Define the average state of the group of N traders as

$$M = \frac{1}{N} \sum_{i=1}^N s_i \quad (3.12)$$

then the definition of the trader's susceptibility χ as

$$\chi = \frac{d(E[M])}{dG} \quad (3.13)$$

Therefore χ controls the level of homogeneity of the trader's positions and as G becomes increasingly positive, global synchronisation emerges leading to wide spread sell-off and the crash of the market.

As a first step, a simple square lattice was considered (like the simple 2-d Ising Model), meaning each trader has four nearest neighbours at the compass points (N,E,S,W). Within the solid state physics analog, there is the critical point at T_c and when $T < T_c$ the coupling order K follows the power law $(K - K_c)^\gamma$ which suggests the relationship

$$\chi \approx A(K_c - K)^{-\gamma} \quad (3.14)$$

where A is a positive constant and $\gamma > 0$ is the *critical exponent* equal to $7/4$ for the 2-d Ising Model. Whilst it is understood that in solid state physics K is driven by the temperature T , the authors do not make any claim as to what the driver of K is during the build up of the financial bubble. But they do make the assumption that its evolution is smooth and analytical around the critical point, although also don't discount for the possibility of it being stochastic so long as standard deviation not too great. They then identify time t_c the first time where $K(t_c) = K_c$ and that prior to this, the evolution of K can be modelled with the following approximation:

$$K_C - K(t) \approx \text{constant} \times (t_c - t) \quad (3.15)$$

then go onto to propose incorporating this approximation in the model for the hazard rate:

$$h(t) \approx B \times (t_c - t)^{-\alpha} \quad (3.16)$$

where B is a positive constant and $0 < \alpha < 1$. Then taking this and combining with (3.6) to get the following description for $p(t)$

$$\log[p(t)] \approx \log[p_c] = \frac{\kappa B}{\beta} (t_c - t)^\beta \quad (3.17)$$

where $\beta = 1 - \alpha \in (0, 1)$ and p_c is the price at the critical time t_c .

The authors make the argument that there are different types of traders (non-professionals, professionals, trading departments in Banks etc) participating within the market, which constitutes a hierarchy and use the q Potts model diamond lattice (see Fig. 2.6) which leads to $\frac{2}{3}(2 + 4^p)$ traders with 4^p relationships (eqn 2.34).

The solution for this, to a first order is [14]

$$\begin{aligned} \chi &\approx \text{Re}[A_0(K_c - K)^{-\gamma} + A_1(K_c - K)^{-\gamma+i\omega} + \dots] \\ &\approx A'_0(K_c - K)^\gamma + A'_1(K_c - K)^\gamma \cos[\omega \log(K_c - K) + \psi] + \dots \end{aligned} \quad (3.18)$$

where A'_0, A'_1, ω, ψ are real numbers and $\text{Re}[Z]$ denotes the real part of a complex number Z . Which shows for the first time the log periodic cycles that increase in frequency as t_c is approached. Now updating the initial simpler hazard rate expressions (see eqn 3.16) to get

$$h(t) \approx B_0(t_c - t)^{-\alpha} + B_1(t_c - t)^{-\alpha} \cos[\omega \log(t_c - t) + \psi'] \quad (3.19)$$

and in the same way refine equation (eqn 3.6) to get the LPPL model

$$\boxed{\log[p(t)] \approx \log[p_c] - \frac{\kappa}{\beta} \{B_0(t_c - t)^\beta + B_1(t_c - t)^\beta \cos[\omega \log(t_c - t) + \phi]\}} \quad (3.20)$$

In a later paper “Large Financial Crashes” [49], Sornette and Johansen refine the JLS model by including the first non-linear correction to cater for a log-frequency shift over time prior to the crash observed by J Feigenbaum and Freund [21].

3.1.1 Non-Linear Generalization of JLS Model

From [49], adopting the renormalization group (RG) approach in respect of the stock market index $I(t)$, it would be related the value of the index at another time t' by the transformations

$$\begin{aligned} x' &= \phi(x) \\ F(x) &= g(x) + \frac{1}{\mu} F(\phi(x)) \quad \text{where } x = t_c - t \text{ and } F(x) = I(t_c) - I(t) \end{aligned} \quad (3.21)$$

where μ is a constant describing the scaling of the index I evolution in respect to a scaling of the time t and $g(x)$ represents the non-singular part of the $F(x)$. Sornette et al provide the following extension, with $\phi(x) = \lambda x$, (eqn 3.1) can be rewritten as

$$\frac{dF(x)}{d \log x} = \alpha F(x) \quad (3.22)$$

and then interpreting (eqn 3.22) as a bifurcation equation F , as a function of a fictitious time ($\log x$) and a control parameter α , with $\alpha = 0$ is the point of bifurcation, which has the complex Ginzburg-Landau expansion (see Sec. 2.1.3)

$$\frac{dF(x)}{d \log x} = (\alpha + i\omega)F(x) + (\eta + i\kappa) |F(x)|^2 F(x) + \mathcal{O}(F^3) \quad (3.23)$$

where $\alpha > 0$, ω , η and κ are all real and letting $F(x) = Be^{i\psi(x)}$ leads to the refined JLS model

$$I(\tau) = A + \frac{(\tau_c - \tau)^\alpha}{(1 + (\frac{\tau_c - \tau}{\Delta t})^{2\alpha})^{1/2}} \left[1 + C \cos(\omega \ln(\tau_c - \tau)) + \frac{\Delta\omega}{2\alpha} \ln \left(1 + \left(\frac{\tau_c - \tau}{\Delta t} \right)^{2\alpha} \right) \right] \quad (3.24)$$

where $\tau = t/\phi$ and $\Delta\omega = B_\infty^2 \kappa$

3.2 Logistic Model for Stock Market Bubbles (top-down)

In their paper “Logistic Model for Stock Market Bubbles and Anti-Bubbles” [33], Lynch and Mestel compare Sornette’s RG based LPPL model with one they derive from a solution to Schrödinger’s equation for the renormalization group (RG). As

before, the RG equation is given by

$$F(x) = \mu^{-1}F(\phi(x)), \quad \phi(x) = \lambda x + O(x^2), \quad x > 0 \quad (3.25)$$

and μ and λ are both positive, $F(0) = 0$ and $F'(0)$ singular. Further assumes $\phi(x)$ and $F(x)$ functions are smooth and continuous at $F(0)$ and that $\phi'(0)$ exists and $\phi(0) = 0$ and $\phi'(0) = \lambda$. Under such conditions, there is a general solution to the RG functional equation with the form

$$F(x) = x^\beta G(\omega \log x), \quad x > 0 \quad (3.26)$$

where $\beta = \log \mu / \log \lambda$, $\omega = 2\pi / \log \lambda$ and $G(x) = G(x + 2\pi)$. But the main goal of their paper was to consider an alternative approach based on Schröder's functional equation and the logistic differential equation

Tackling the general case of a non-linear $\phi(x)$, they propose a change of x -coordinate and make the assumption that there is a solution $\psi(x)$ around $x = 0$ for the Schröder equation $\psi(\phi(x)) = \lambda \psi(x)$ also satisfying $\psi(0) = 0$. Under these conditions, $\psi(x)$ conjugates $\phi(x)$ to its linearized form λx around $x = 0$. Therefore under this change of x -coordinate, the Schröder functional equation and its general solution become

$$\begin{aligned} \tilde{F}(\psi(x)) &= \mu^{-1} \tilde{F}(\lambda \psi(x)), \quad \psi(x) > 0 \\ \tilde{F}(\psi(x)) &= \psi(x)^\beta G(\omega \log \psi(x)), \quad x > 0 \end{aligned} \quad (3.27)$$

where

$$\begin{aligned} \beta &= \frac{\log \mu}{\log \lambda} \\ \omega &= \frac{2\pi}{\log \omega} \end{aligned}$$

If $\phi(x)$ is the time-1 map of an autonomous differential equation (see Sec. 2.3.6), then the conjugating function $\psi(x)$ can be found and thereby provide a close form solution for the Schröder's functional equation.

$$\frac{dx}{dt} = \rho h(x), \quad x(0) = x_0, \quad h(x) = x + O(x^2) \quad (3.28)$$

solving by separation of variables, define a new function

$$H(x(t)) = \int \frac{dx}{h(x)} = \rho t + C$$

$$H(x(t)) - H(x_0) = \rho t$$

where C is the constant of integration and $x_0 = x(0)$. Then by exponentiating both sides to get

$$e^{H(x(t))} = e^{H(x_0)} e^{\rho t} \quad (3.29)$$

and then making the following definitions

$$\begin{aligned} \phi(x_0) &= x(1) \\ \psi(x_0) &= e^{H(x_0)} \\ \lambda &= e^\rho \end{aligned} \quad (3.30)$$

we see that (3.29) has the form of Schröder's functional equation

$$\psi(\phi(x_0)) = \lambda \psi(x_0)$$

Now consider the special case of the logistic differential equation given by

$$\frac{dx}{dt} = \rho x(1 + \nu x), \quad x(0) = x_0, \quad \rho = \log \lambda, \quad \lambda > 1 \quad (3.31)$$

has the solution

$$x(t) = \frac{x_0 e^{\rho t}}{1 + \nu x_0 (1 - e^{\rho t})} \quad (3.32)$$

and then insert the definitions (3.30) to get the fractional linear time-1 map (see Sec. 2.3.6)

$$\phi(x_0) = \frac{\lambda x_0}{1 + \nu x_0 (1 - \lambda)} \quad (3.33)$$

and then finally replacing x_0 with x to get the non-linear perturbation of $\phi(x) = \lambda x$, parameterised by ν

$$\phi(x) = \frac{\lambda x}{1 + \nu x (1 - \lambda)} \quad (3.34)$$

and the conjugating function $\psi(x)$

$$\psi(x) = \frac{x}{1 + \nu x} \quad (3.35)$$

and then by re-writing $\nu = 1/\Delta_t$ to get a form comparable with Sornette (see Sec. 3.1.1)

$$\begin{aligned} I(t) &= A + \frac{|t_c - t|^\beta}{\left(1 + \frac{|t_c - t|}{\Delta t}\right)^\beta} [B + C_1 \cos \theta(t) + C_2 \sin \theta(t)] \\ \theta(t) &= \omega \ln |t_c - t| - \omega \ln \left(1 + \frac{|t_c - t|}{\Delta t}\right) \end{aligned} \quad (3.36)$$

as required.

CHAPTER 4

SUMMARY OF (TOP-DOWN) AND (BOTTOM-UP) APPROACHES

Within the study of Physics, researchers and scientists often adopt the approach of *methodological reductionism* [10] , or what this paper refers to as the “bottom-up” approach. In other words the desire to provide an explanation for some observed phenomena in terms of smaller constituent parts. In the study of Thermodynamics or Quantum Mechanics, this would require the modelling of individual atoms and molecules. In the former this poses unsurmountable problems due to the large numbers involved ($> 10^{23}$) and in the latter, due to fundamental constraints such as the Uncertainty Principle [26]. Explanations for the macroscopic observables cannot be given purely in terms of the microscopic constituents. The classical physics approach to Thermodynamics is to make some simplifying assumptions around the state of the system being in some sort of equilibrium in respect to temperature and energy and treat the whole ensemble in probabilistic way [42]. Thus was born the theory of Statistical Mechanics, pioneered by Josiah Willard Gibbs who coined the name *statistical mechanics*, James Clerk Maxwell, Rudolf Clausius and Ludwig Boltzmann (see appendix A). The dilemma was that whilst *classical* thermodynamics could make good predictions and models for certain macroscopic measurements such as energy, entropy and specific heat, all of which were based on the assumption that inter molecular interactions could be ignored; other types of phenomena such as critical points at phase transition did require the consideration of inter molecular forces. The solution to this problem came in a number of ways (1) van der Waal’s forces between atoms, (2) Ising’s

simplified lattice model for atomic arrangement, (3) Weiss's molecular field theory, (4) Landau's theory of λ -point transitions, (5) renormalization group theory first developed in quantum field theory and (6) Wilson's application of renormalization group to solid state physics and thermodynamics. Landau introduced the idea of an *order parameter* which was a measure of how ordered the system was before and after the critical temperature of a phase transition (see Sec. 2.1.4). Landau also stressed the importance of symmetry (in relation to degrees of freedom) near the point of transition. In 1925 Lenz proposed a very simple atomic model based on a lattice to his doctoral student Ising. Where each spin located on a lattice node can align itself with or against an external magnetic field and the question as to whether this model was capable of capturing spontaneous magnetization at any non-zero temperature. Ising was able to find a solution to the one dimensional lattice but this was too simple to provide spontaneous magnetization. But the concept of the lattice model (generally referred to now as the Ising model) has proven a very powerful abstraction for many different fields of physics, computer science and mathematics to this day. In 1944 Lars Onsager published an exact solution [37] for the two dimensional Ising model and in the process delivered a knock-out blow to classical statistical mechanics by showing amongst other things, that the partition function was non-analytic at the critical temperature T_c which meant that the Landau expansion was invalid near the critical points (see Sec. 2.1). Then in 1949, Onsager calculated that the spontaneous magnetization had the form $-(\tau)^{1/8}$ near the critical point which was very different to the form $(-\tau)^{1/2}$ predicted by the Weiss molecular field theory.

In the early 1960s, people were thinking about these different exponents characterising the various thermodynamic properties and asking if there was any relationship between them (see Sec. 2.3.2).

In 1966 Kadanoff noted that near T_c , the correlation length ξ becomes very large in comparison with the lattice spacing L in the Ising model and consequently took the approach of replacing individual spins with the average of a block of spins (see sec 2.1.5) as a means of addressing the change of scale near the critical point. So

given the current situation people were looking for ways to reconcile the two different groups that were either continuous or discontinuous with a type of *universality* for short range forces with the general form given by (eqn 2.54). In 1971, Kadanoff collected all the critical exponents into a *hypothesis of universality*¹

In the early 1970s K.G. Wilson took the renormalization group (RG) approach developed by Stueckelberg & Peterman [38] and then independently by Gell-Mann & Low [22] for quantum field theory with a view to using it to model *universality*, building on the Kadanoff's block spin model. The RG is a transformation $\mathbf{R}$ of the original Hamiltonian $\mathcal{H}$ with N degrees of freedom into a new Hamiltonian $\mathcal{H}'$

$$\mathcal{H}' = \mathbf{R}[\mathcal{H}] \quad (4.1)$$

with a reduction in degrees of freedom N' whilst chosen to preserve the partition function $Z_{N'}(\mathcal{H}') = Z_N(\mathcal{H})$. Kadanoff's block spin is one of a number of ways to reduce the number of internal (spin) interactions. An important feature of the transformation is that it is iterative

$$\mathcal{H}^{n+1} = \mathbf{R}[\mathcal{H}^n] \quad (4.2)$$

and the fixed points of the repeated iterations provide the universality [58] and each fixed point corresponds to a different universality class. The RG flow provides a different view of phase transition and critical phenomena through the study of these iteration maps that act on the scaled Hamiltonians. Each iteration of the RG map reduces the degrees of freedom within the Hamiltonian of the partition function. An attractive fixed point represents a thermodynamic phase and a hyperbolic fixed point represents a phase transition. The critical exponents are connected by simple relations to the linearised map near the hyperbolic fixed points. The RG approach has been very succesful in providing a mathematical framework for the scale invariance and universality of the critical exponents. The predicted values from RG have have proven to be in very good agreement with precise measurements made aboard the space shuttle [32].

¹patterns of behaviour that recur and repeat across different physical systems [8]

Surprisingly, the reductionist approach was unable to provide an explanation for the macroscopic behaviours in terms of the microscopic constituents as the correlation length $\xi \rightarrow \infty$ and needed Wilson to provide the renormalization *bridge* between the reductionist's *bottom-up* and the emergent *top-down*. Whilst the RG predictions have been proven to match experimental observations, it leaves the question of a causal explanation. Margaret Morrison, in her paper “The Non-Causal Character of Renormalization Group Explanations” [36] makes the argument that the observed universal features of the critical exponents cannot be ex-

plained in terms of system's interacting parts. Instead it is the role of the fixed points of the RG flows together with the way RG acts on the Hamiltonians (see Fig. 4.1). The iterative application of the RG map (eqn 2.25) is related to a scale invariance symmetry which shows how the system appears the same at all scales, given a particular value in the associated order parameter (see Sec. 2.1.4).

Sornette et al, in their paper “Stock Market Crashes, Precursors and Replicas” [50] make the argument that market crashes have identifying precursory patterns that suggest a kind of dynamical critical point with a characteristic log-periodic signature, similar to those found in seismological data and acoustic data of ruptures in pressurised tanks. Another feature of the data sets is a discrete scale invariance, which is the hallmark the log periodic power law and the equation they propose to model the data (see eqn 3.1). The paper then makes a claim for a 4-parameter fit, where the objective function is to minimise var_{lp} and further propose the variables m and ω could capture some universality equivalent to the uni-

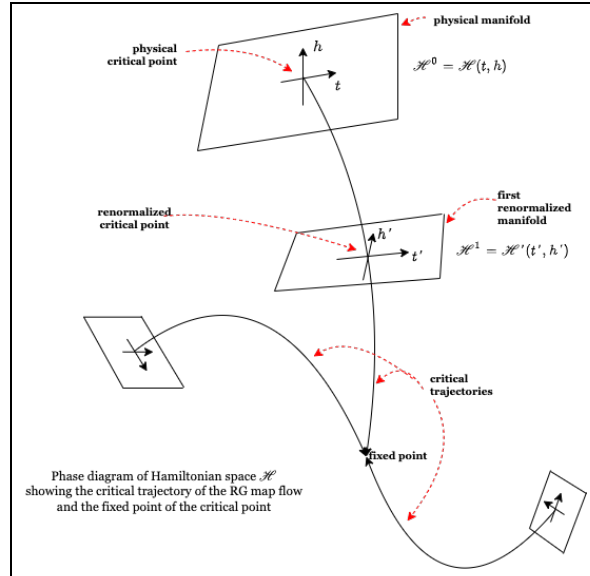


Figure 4.1: Hamiltonian space phase diagram

versality classes predicted by renormalization group flows. They use the equivalent of the thermodynamic reduced temperature (eqn 2.11) in respect of measuring time relative to the time of the critical event ($t_c - t$). In a later paper “Crashes as Critical Points” [29] (see Sec. 3.1), A Johansen, O Ledoit and D Sornette develop a concept of “noisy traders” that imitate, in a herd-like behaviour [2], where they can be in one of two states *buying*, *selling*, which is an intentional analog with spin states in the Ising Model (see Sec. B) and the associated iterated maps that describe the RG. Where they further develop the argument that the mass sell-off, that precipitates the crash is at a critical point and can be modelled using the RG approach Wilson developed for the Potts Model and now commonly referred to as the JLS model. As before they also make the argument for a kind of emergent behaviour from a self-organized critical system [7] [6]. Both papers claim they provide empirical evidence for successfully fitting the log periodic power law model to the data and a predictive power.

In their 1995 paper “Discrete Scaling in Stock Markets Before Crashes” [21], J Feigenbaum and Freund supported (independently) the idea of stock market crashes as critical points in a hierarchical system with discrete scaling “in the spirit of the known earthquake-stock market analogy” and also claiming evidence for a log-periodicity fit. But in later papers [20] [11], Feigenbaum and Chang go onto question the idea of log periodic precursors and viewing the martingale constraint (eqn 3.2) used in developing the JLS model, as overly restrictive when modelling rationality of trading

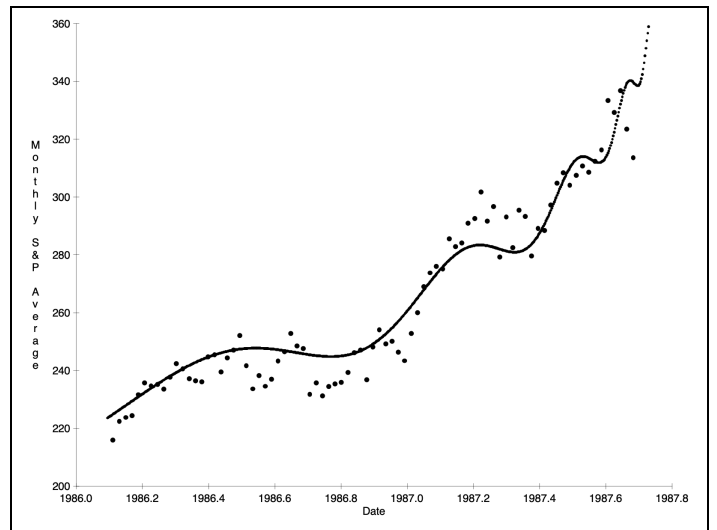


Figure 4.2: J Feigenbaum Freund [21] Fitting LPPL Model to S&P Market Data

$$C = -0.035, \alpha = 0.2, \omega = 8.06, \\ t_c = 87/10/9$$

agents. Feigenbaum does not refute the hypothesis but no longer finds log periodic component statistically significant and also questions the efficacy of standard statistical measures in testing hypothesis for LPPL. Concluding, that to advance such a hypothesis, researchers need to agree an objective definition of what constitutes a log-periodic window. Further noting that the economic agents (market events, trader sentiment etc) have far more unknowns and variation compared with measurements taken from a typical physics experiment involving statistical mechanics. The debate over LPPL has continued over the past 20 years and Sornette and Johansen published another paper addressing the various criticisms [51]. Sornette and Johansen extended their JLS model to include a first non-linear correction, by incorporating a complex ginzburg-landau expansion [49] (see Sec 3.1.1 & Sec. 2.1.3), to account for the log-frequency shift noted by J Feigenbaum [21]. The authors also claim an improvement in the earlier detection of bubbles forming.

In their paper “Renormalization Group Functional Equations” [13], Curtright and Zachos focus on the functional conjugation methods used in Gell-Mann and Low’s paper [22] and make them first class citizens in the study of renormalization group and in doing so, gaining insight into the interplay between continuous and discrete scaling (see Sec. 2.3.6).

Following Sornette and co-workers development of the LPPL JLS Model, which was inspired by the Potts Model and Wilson’s RG work; Lynch and Mestel (see Sec. 3.2), apply the aforementioned Curtright and Zachos paper, to develop an approach based on Schröder’s equation to generate a LPPL model for stock market prices.

While the authors make no claim, their *logistic differential equation* generates a more accurate LPPL based model, they show, under certain conditions, it is as good as the extended JLS model (eqn. 3.24). But have introduced a novel construction of a suite of log-periodic power-law models for asset bubbles from the solutions to Schröder’s equation (eqn 2.52) and without recourse to the machinery of Ising Models and Statistical Mechanics; which is why this paper views this as

an example of a “top-down” approach in contrast to the JLS “bottom-up”.

Both models seem to face the same challenge of how to fit the LPPL model parameters to the data and there is a sensitivity to both the chosen window size, when the window starts and agreement on the selection of the critical date t_c . Some researchers have claimed LPPL models are intrinsically hard to fit to time series, due to the squared residuals being very sensitive to certain combinations of the parameters [9].

Claims have been made as to the universality of LPPL model [4] [18], but this is not widely accepted and lacks the evidence that is available to support the universality classes in the physical world. Nonetheless, despite the ongoing debates, LPPL models are still an active area of research.

CHAPTER 5

CONCLUSION

Despite the brilliant work and early efforts of Maxwell, Clausius, Landau and others, the resultant mean-field theory, so successful in many ways, nonetheless was unable to account for interplay at the microscopic and macroscopic close the critical point. It took Kadanoff and finally Wilson to use the renormalization group developed to tame the divergent series in quantum electrodynamics, to account for the symmetries and scale-invariance at the critical point. Within Gell-Mann and Low's RG formulation, was Schröder's functional equation, which provided the scale invariance via functional conjugation. This was also the link to the world of dynamical systems and universality classes.

The initial JLS model was predicated on the Potts Model representing relationships between traders but suffered with log-frequency shift which they corrected by incorporating the Ginzburg-Landau amplitude equation, series expansion. An equation associated with turbulent flow and thus a mathematical link to discrete scale invariance. Lynch and Mestel's logistic differential equation provided an alternate route to the log-periodic power-law, that could give a good account of the bubble's growth and crash. This did not require consideration of an underlying hierarchy such as the Potts Model.

A recurring theme has been one of rescaling of coupling parameters between interacting agents, be they atoms or traders. Which exhibit a scale invariance through appropriate power laws and symmetry constraints. In recent years there has been a revival of Geoffrey Chew's work known as *bootstrapping* that envisioned a self-determining universe that required all particles be equally composite and funda-

mental [39]. There is now a lot of interest in developing his ideas under the name of conformal field theory; with conformal invariance being a feature of the fixed point theory within renormalization group and providing a deeper insight into the rescaling of the microscopic world.

APPENDIX A

THERMODYNAMICS AND STATISTICAL MECHANICS

Here we provide a quick overview of the elements of statistical mechanics referenced in this paper [34].

Statistical mechanics is a branch of physics that provides models for the energy E and entropy S of very large numbers N of identical particles (typically of the order $\approx 10^{23}$) contained within a volume V and at temperature T creating a pressure P dependent on the type of material.

This assembly of particles, constrained by P, T, V will present at the macroscopic level observable (measurable) thermodynamic properties and these variables provide a specification of the assembly at the *macrostate*.

This assembly can also be given a specification at the microscopic level by considering (in the classical sense) positions and velocities of each of the individual particles. For a three dimensional volume the *microstate* will have $6N$ degrees of freedom all of which will be changing from one instant to the next. Clearly to try and calculate such a large number of variables ($(6N)^{23}$) would not be possible and therefore the microscopic description must be a statistical one and this is done based on two basic postulates. The first is that under equilibrium all possible microstates are equally probable and second, assume that Boltzmann's definition of entropy is equivalent to the the usual thermodynamic entropy. Mathematically, these two postulates are stated as

$$\begin{aligned}
p(n_s) &= 1/\Omega, \text{ where} \\
\sum_s n_s &= N \\
S &= k \ln \Omega
\end{aligned} \tag{A.1}$$

where n_s are the number of particles at a given energy state s , Ω is the total number of possible equiprobable energy states and k is the Boltzmann constant. Adherence to these two postulates provides a consistent and successful bridge between the microscopic and macroscopic descriptions of matter, which is stated as

$$F = -kT \ln Z \tag{A.2}$$

where F is the Helmholtz free energy

$$F = E - TS \tag{A.3}$$

and where Z is the partition function

$$\begin{aligned}
Z &= \sum_i \exp[-E_i/kT] \\
&= \sum_i \exp[-\beta E_i] \text{ where } \beta = \frac{1}{kT}
\end{aligned} \tag{A.4}$$

where i is the i^{th} particle. The partition function can then be used for calculating various statistical quantities for the energy E .

$$\begin{aligned}
\overline{E} &= -\frac{\partial}{\partial \beta} \ln Z \\
\overline{E^2} &= \frac{1}{Z} \frac{\partial^2 Z}{\partial \beta^2} \\
\sigma^2 &= \overline{E^2} - \overline{E}^2 \\
&= \frac{\partial^2}{\partial \beta^2} \ln Z
\end{aligned} \tag{A.5}$$

The differential form of the Helmholtz free energy A.3 is given by

$$dF = dE - T dS - S dT \quad (\text{A.6})$$

then using the first law for dE to get the differential form

$$dF = -S dT - p dV + \mu dN \quad (\text{A.7})$$

then the average differential is given by

$$d\bar{F} = -\bar{S} dT - \bar{p} dV + \bar{\mu} dN \quad (\text{A.8})$$

leading to the following thermodynamic laws

$$\bar{S} = - \left(\frac{\partial \bar{F}}{\partial T} \right)_{V,N}, \quad \bar{p} = - \left(\frac{\partial \bar{F}}{\partial V} \right)_{T,N}, \quad \bar{\mu} = - \left(\frac{\partial \bar{F}}{\partial N} \right)_{T,V} \quad (\text{A.9})$$

Mathematically, the critical point in thermodynamics relating to specific heat capacity for example, will be expressed as a discontinuity in one of these partial derivatives.

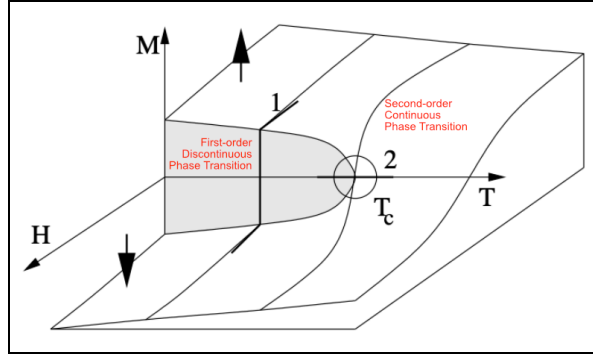


Figure A.1: Phase Transition of an Ising Ferromagnetic

There are two ways the magnetization of a ferromagnet can change. In the first case labelled (1) in Figure A.1, at $T < T_c$ the two spin states (*up/down*) coexist and there is a discontinuity at $H = 0$ and is a first-order phase transition. The correlation length for a first-order transition is generally finite. In the second case (2) at $T \geq T_c$, the transition is continuous and the correlation length becomes infinite and is said to be a second-order transition. Therefore as the critical point is approached from either side, the correlation length diverges.

APPENDIX B

ISING MODEL HAMILTONIAN

In order to study phase transitions and critical phenomena, various models have been developed over the years, each a refinement on the last. The first of these considered here is the Ising Model, which is a regularly spaced d -dimensional lattice with a regular spacing of the nodes. Here we consider a two-dimensional regular lattice array.

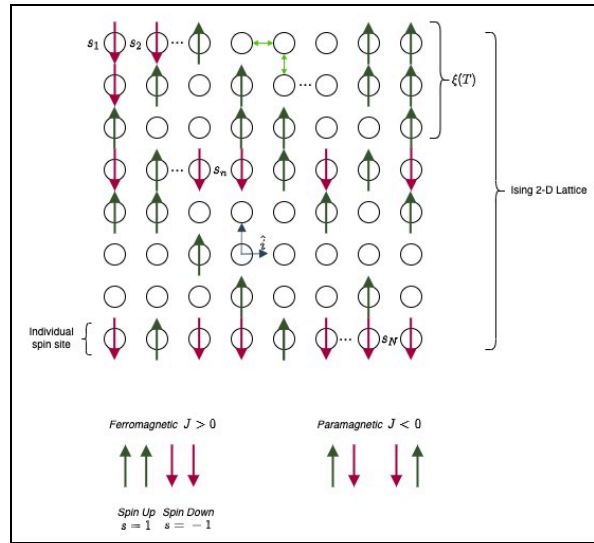


Figure B.1: Ising Square Lattice Models

Each node represents the spin of an atom (see Figure B.1), which is a quantum mechanical representation of a magnetic dipole and its value is constrained to the set $\{-1, 1\}$, where the up arrow represents spin value 1 and a down arrow repre-

sents the spin value -1.

$$s_i \in \{-1, 1\}$$

The total energy of the collection of spins $\{s_i\}$ is given by

$$E = -J \sum_{\langle i,j \rangle} s_i s_j - B \sum_i s_i \quad (\text{B.1})$$

Where the first term J is due the level of interaction between neighbouring spins and how they like to be aligned and the second term is due to an external magnetic field B .

If $J > 0$, then neighbouring spins prefer to be aligned the same way either as $\uparrow\uparrow$ or $\downarrow\downarrow$ and such a system is referred to as *ferromagnetic*.

If $J < 0$, then neighbouring spins prefer to be aligned in opposition $\uparrow\downarrow$ and is *anti-ferromagnetic* or *paramagnetic*.

The third ingredient is Temperature T and the effect this has on the individual spins s_i and their interactions $s_i s_j$. For ferromagnetic materials, the lower energy state is achieved when the individual spins align with the external magnetic field B and when the interaction energy J is positive this further encourages the spins to be all aligned the same way (see Figure B.1). This alignment of the spins creates what we would measure at the macroscopic scale as magnetization M . But as the Temperature T increases, this provides energy for the spins to overcome the influences of B and J . As the spins move away from this organised state at lower temperatures, the entropy of the system S , increases.

So as the temperature increases, the number of different spin states increases.

Where a spin state is a particular configuration of all the individual spin states of up or down $\{s_i\}$. If the lattice consists of N sites then there are 2^N possible spin state configurations.

In the canonical ensemble [34], the probability of being in a given state $\{s_i\}$ is

given by

$$p(s_i) = \frac{e^{-\beta e(s_i)}}{Z} \quad (\text{B.2})$$

where $\beta = 1/T$ and the normalization factor Z is called the partition function

$$Z(T, J, B) = \sum_{\{s_i\}} e^{-\beta E[s_i]} \quad (\text{B.3})$$

The average spin of the configuration, known as the equilibrium magnetization is given by

$$M = \frac{1}{N} \left\langle \sum_i s_i \right\rangle, \text{ where } m \in [-1, +1] \quad (\text{B.4})$$

The Hamiltonian for the Ising Model in a magnetic field is given by

$$H = -J \sum_n \sum_i s_n s_{n+\hat{i}} + \mu B \sum_n s_n \quad (\text{B.5})$$

where the $\{\hat{i}\}$ are the unit vectors for each axis of the lattice and μ is the gyromagnetic ratio. The thermodynamics can be obtained from the partition function

$$Z = \sum_{\{s_i\}} e^{-H/kT} \quad (\text{B.6})$$

where the sum is over all possible spin configurations $\{s_i\}$ and k is Boltzmann's constant. The free energy F is given by

$$F = -kT \ln Z \quad (\text{B.7})$$

The magnetization per lattice site

$$M = \frac{1}{N} \sum_n \frac{\langle \mu s_n e^{-H/kT} \rangle}{\langle e^{-H/kT} \rangle} \quad (\text{B.8})$$

where $\langle \dots \rangle$ denotes the summing over all spin configurations

$$M = -\frac{1}{N} \frac{\partial}{\partial B} \left(\frac{F}{kT} \right) \quad (\text{B.9})$$

Spontaneous magnetization can occur in ferromagnetic materials, due to an ordered spin state without an external magnetic field, when their temperatures are $T \leq T_c$, where T_c is known as the critical point, which in the case of magnetization is also known as the Curie Temperature.

This spontaneous magnetization is related to the correlation between the different spin sites and the strength of this correlation is given by the correlation function

$$\Gamma_n = \frac{\langle s_n s_0 e^{-H/kT} \rangle}{\langle e^{-H/kT} \rangle} \quad (\text{B.10})$$

At $T = 0$ all the spins are aligned giving $\Gamma_n = 1$ and independent of n . As $T \rightarrow \infty$, then $\Gamma_n(T, n) \rightarrow 0$ but is dependent on n and T_c . The Ising lattice contains two lengths of interest, (1) the lattice spacing l and (2) the correlation length $\xi(l)$, which is the length over which there is correlation between Ising sites. As we shall show, $\xi(l) \rightarrow \infty$ as $T \rightarrow T_c$.

The purpose of the Ising Model is to provide a phenomenological based explanation for these critical points and the various macroscopic thermodynamic values. For a real material, the vast number of atoms, the available degrees of freedom is of the order $\approx 10^{23}$. In contrast, the simple Ising lattice structure with the additional constraints on the allowed orientation of the spins is a gross simplification of the real world, but nonetheless solutions to the Ising Model have been successful in capturing thermodynamic behaviours and also these critical phenomena.

The early solutions to the Ising model only require classical statistical physics, although later work has used quantum field theory. Indeed the development of renormalization methods to remove singularities in quantum mechanics was the inspiration for Wilson (working from Kadanoff's block model) in developing the Renormalization Group approach to solving the Ising Model and successfully capturing the critical phenomena.

At temperatures $T > T_c$, the correlation length ξ is of the order of atomic distances between neighbouring atoms. But at the critical point T_c , where phase transition occurs, then $\xi \rightarrow \infty$, meaning that spin alignment extends across

macroscopic scales. These interactions, fluctuate across space and time and are the key ingredient to capturing the macroscopic properties of the phase change. As well as these discontinuities and singularities at the critical point, another hallmark of the spin fluctuations are that they have fractal properties of self-similarity across scales ranging from the microscopic to the visible macroscopic. And of course self-similarity is the hallmark of the power law. An example of these can be seen at the transition point from liquid to gas where at the critical point both phases exist and produce an opalescent vapour quite distinct from the liquid or gas.

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